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|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMPANY  | McKinsey & Company                                                                                                                                                                                  |
| ROLE     | Senior Data Scientist - Scientific AI                                                                                                                                                               |
| POSTING  | <a href="https://www.mckinsey.com/careers/search-jobs/jobs/seniordata scientist-scientificai-106244">https://www.mckinsey.com/careers/search-jobs/jobs/seniordata scientist-scientificai-106244</a> |
| TO       | Scientific AI Hiring Team                                                                                                                                                                           |
| LOCATION | Boston   Chicago   New Jersey   New York City   Raleigh + 3 more                                                                                                                                    |
| DATE     | 2026-07-10                                                                                                                                                                                          |

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Dear Scientific AI Hiring Team,

I am applying for the Senior Data Scientist role on McKinsey's Scientific AI team within the Life Sciences practice. Building SNPTX - a governed execution layer for multi-modal biomedical ML - has been an unbroken four-year exercise in precisely the intersection your team names: PyTorch model development across drug-discovery, multi-omics, and clinical time-series data; autonomous experimentation grounded in Bayesian optimization; LangChain agent tooling; and delivery through calibration-corrected, reviewer-grade benchmarks.

The load-bearing artifact is snptx-repro-chembl, a five-minute end-to-end reproduction of a ChEMBL bioactivity graph convolutional network baseline that runs from a clean clone on commodity CPU (accuracy 0.9851, 98.0% reproducibility under fixed seeds). It is one of two bounded reproductions inside SNPTX; the other is snptx-repro-deepdds, a bounded reproduction of the DeepDDS drug-synergy graph-attention model. Both use PyTorch, both are inspectable end-to-end without follow-up, and both feed the broader benchmark surface that reports 99.3% imaging, 92.8% omics, 98.5% drug-GCN, 97.5% ChEMBL antibacterial 5-fold CV, and 93.0% Visium spatial-transcriptomics fusion on declared configurations. The signal I care about is not the top-line metric - it is that a second party can re-execute the pipeline and get the same number.

The autonomous surface is where I would foreground the fit against your ML-methods list. SNPTX's `ExperimentEngine` runs Gaussian-process surrogates with Expected-Improvement + Value-of-Information acquisition and sequential-probability-ratio-test stopping ( $\alpha = \beta = 0.05$ ), backed by a DuckDB experiment catalog; validated throughput is 1,037 experiments per hour with 104 discoveries per 1,100-experiment campaign (surrogate ECE = 0.055). That covers Bayesian optimization, surrogate models, calibration, and design-of-experiments in one working component, and gives me a concrete answer for the longitudinal and survival-adjacent side of your list too: the CSCI E-116 final project on eICU ICU-mortality forecasting (VAR, dynamic-factor, state-space, decomposition) routed straight into SNPTX under the same deterministic harness.

Concretely against the qualifications you list: (1) Python across pandas, numpy, scikit-learn, pytorch, and langchain, in production across SNPTX and FIVSS. (2) Deep-learning breadth - graph neural networks (ChEMBL GCN, DeepDDS graph-attention), diffusion, and transformer-based NLP including BiLSTM-CRF and BERT-family biomedical NER on BC5CDR. (3) Agents / LangChain / prompt orchestration built into SNPTX. (4) Knowledge graphs via Hetionet and Open Targets with NetworkX and Neo4j analytics over ChEMBL32. (5) Life-sciences and therapeutics context across eight modality families and 46 datasets

(ChEMBL, Hetionet, Open Targets, TCGA, GTEx, MIMIC-IV, AlphaFold, Visium, DepMap, BC5CDR, eICU).  
(6) Production code and object-oriented ownership demonstrated by ten years running FIVSS as sole founding engineer through acquisition, and by SNPTX's engineering discipline (1,357 tests, 0 Ruff, 0 Pyright). Computer vision remains an honest gap; I would grow into it rather than claim mastery today.

I can start immediately and am authorized to work in the United States without sponsorship. The reviewer surface I would ask you to open first is <https://snptx1.github.io/snptx-academic/index.html#for-review> - three short reading paths, an inspection-invariants panel, and direct routes to the two reproduction repositories and the four capability surfaces.

Sincerely,

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